

Hunter Regional Readiness Report



A National Action Plan
Case Study



MAY 2025

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Acknowledgements

Beyond Zero Emissions Project Team

Strategic Oversight: Heidi Lee

Authors: Dr Heidi Edmonds, Tejesh Kashyap, Sam Mella, Beth Mitchell, Kelvin Wicks

Staff contributors: Michelle Lee, Adam Waryszak, Carissa Dyall, Dr Anna Boin, Dr Fred Hiskens

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About this Compendium

Preface

This document is a **compendium to Beyond Zero Emissions' (BZE) *Powering Up the Hunter***, which **provides a summary of findings** based on our research on the Hunter, and should be read in conjunction with the summary.

This compendium provides a detailed assessment based on a **subset of the National Action Plan's Critical Success Criteria used to assess progress in the Hunter region**. The selected criteria (shaded in blue in Table 1) **evaluate the supply and demand for renewable energy in the region, the readiness of key infrastructure, as well as the government policies and the industry commitments that shape them**. The subset serves as a foundation for future research and collaboration to assess the broader Critical Success Criteria across the Hunter region.

Table 1. National Action Plan themes and Critical Success Criteria

 Enabling Infrastructure	 Industry Capability & Capacity	 Policy & Governance	 Social Acceptance	 Financing the Transition
Energy System Renewable energy generation Transmission network Distribution network Energy storage Hydrogen network Energy system orchestration	Industry Decarbonisation Industry decarbonisation plans Workforce Capacity to Deliver Existing workforce capacity Reskilling and upskilling programs	Federal Policy Emissions reduction commitment Skills and training Approvals processes Investment framework Research and development Government procurement support	Engagement & Participation of Communities Knowledge sharing and co-design forums Effective First Nations engagement Effective community engagement Public awareness and education	Investment Public finance commitment Attracting private investment
Transport System Port Rail Road Transport networks	Innovation & Knowledge Development Innovation Commercialisation Knowledge generation and sharing	State Policy Emissions reduction commitment Skills and training Approvals processes Investment framework Research and development Government procurement support	Community Benefits & Impacts First Nations community benefit sharing Wider community benefit sharing Strategic and nature positive land-use planning Demographic planning	Risk & Resilience Risk and resilience
Water System & Waste Management Water supply Wastewater management Solid waste & resource recovery	Collaboration & Industrial Ecosystems Industrial symbiosis & material exchange Coordinated infrastructure plan for common user infrastructure Contribution to Australian supply chain resilience International linkages / international coordination	Local Government Policy LGA decarbonisation plans Regional climate adaptation/ resilience plan		
Social Infrastructure Housing Health Education		Governance Regional coordination Regional representation of key agencies Policy alignment and coordination		
Communications Infrastructure Digital infrastructure				

 Evaluated as part of preliminary assessments
 Criteria yet to be investigated

Details on the process of assessment for each criteria can be found in the [National Action Plan: Regional Assessment Process \(Version 1\)](#).

Out of scope

Social Acceptance and Common Use Infrastructure: Transparent communication of development processes and clear communication of benefits are key to building local support for large infrastructure projects. This extends beyond renewable energy infrastructure to include all common user infrastructure and community infrastructure vital for regional development such as roads, water and waste infrastructure, and health and education services.

Although a detailed assessment of these criteria and social acceptance was outside of scope of the *Powering up the Hunter* and the *Hunter Regional Readiness Report*, we acknowledge their importance in fostering social engagement and support as critical to the successful delivery of all new infrastructure projects.

Accounting for Coal Emissions and Closures: Similarly, there is a need to better understand the relationship between coal mine closures, fossil fuel emissions, and the progress of cleantech.

While these elements fall outside BZE's usual remit, **we are inviting researchers and community groups to partner with us** to assess these important pieces of the puzzle.

Overview

The Hunter Critical Success Criteria Assessment is the first detailed assessment in a series that will be conducted across Australia's industrial regions and clean economy opportunities.

Insights from this and future assessments will inform the National Action Plan—a master plan to efficiently unlock Australia's zero-emissions economy and drive long-term prosperity in industrial regions.

The **National Action Plan will define a clear sequence of the 'what', 'where', and 'when' major projects must happen**, so that Australia retains our significant role in global energy markets, while laying the foundations for a resilient and sustainable zero-emissions economy.

This body of work is different from other reports published by BZE. It is a work in progress and invites new partners, collaborators, and stakeholders to join us as we build the National Action Plan.

The project began in July 2024 with the engagement of industry, government, community and research stakeholders in the identification of key themes and Critical Success Criteria to assess regional progress toward successful coordination of Australia's clean energy and industry opportunity (Table 1).

By testing our strategies and solutions in the complex, high-stakes environment of the Hunter region, we aim to provide scalable insights that can drive meaningful change both regionally and nationally. The assessment provides the **state of play in the region as of April 2025**. An overview of the assessment is presented in Table 2. The *Powering Up the Hunter* report provides a summary of findings and recommendations from this assessment.

Table 2. Summary of Critical Success Criteria Assessment

1. Enabling Infrastructure	2. Industry Capability & Capacity	3. Policy & Governance	4. Social Acceptance	5. Financing the Transition
1.1 Energy System	2.1 Industry Decarbonisation	3.1 Federal Policy	4.1 Engagement & Participation of Communities	5.1 Investment
1.1.1 Renewable energy generation	2.1.1 Industry decarbonisation plans	3.1.1 Emissions reduction commitment	4.1.1 Knowledge sharing and co-design forums	5.1.1 Public finance commitment
1.1.2 Transmission network	2.2 Workforce Capacity to Deliver	3.1.2 Skills and training	4.1.2 Effective First Nations engagement	5.1.2 Attracting private investment
1.1.3 Distribution network	2.2.1 Existing workforce capacity	3.1.3 Approvals processes	4.1.3 Effective community engagement	5.2 Risk & Resilience
1.1.4 Energy storage	2.2.2 Reskilling and upskilling programs	3.1.4 Investment framework	4.1.4 Public awareness and education	5.2.1 Risk and resilience
1.1.5 Hydrogen network	2.3 Innovation & Knowledge Development	3.1.5 Research and development	4.2 Community Benefits & Impacts	
1.1.6 Energy System Orchestration	2.3.1 Innovation	3.1.6 Government procurement support	4.2.1 First Nations community benefit sharing	
1.2 Transport System	2.3.2 Commercialisation	3.2 State Policy	4.2.2 Wider community benefit sharing	
1.2.1 Port	2.3.3 Knowledge generation and sharing	3.2.1 Emissions reduction commitment	4.2.3 Strategic & nature positive land-use planning	
1.2.2 Rail	2.4 Collaboration & Industrial Ecosystems	3.2.2 Skills and training	4.2.4 Demographic planning	
1.2.3 Road	2.4.1 Industrial symbiosis & material exchange	3.2.3 Approvals processes		
1.2.4 Transport	2.4.2 Coordinated plan for common user infrastructure	3.2.4 Investment framework		
1.3 Water System & Waste Management	2.4.3 Contribution to Australian supply chain resilience	3.2.5 Research and development		
1.3.1 Water supply	2.4.4 International linkages / coordination	3.2.6 Government procurement support		
1.3.2 Wastewater management		3.3 Local Government Policy		
1.3.3 Solid waste & resource recovery		3.3.1 LGA decarbonisation plans		
1.4 Social infrastructure		3.3.2 Regional climate adaptation/ resilience plans		
1.4.1 Housing		3.4 Governance		
1.4.2 Health		3.4.1 Regional coordination		
1.4.3 Education		3.4.2 Regional representation of key agencies		
1.5 Communications infrastructure		3.4.3 Policy alignment and coordination		
1.5.1 Digital infrastructure				

Legend

5 (Very High)
4 (High)
3 (Medium)
2 (Low)
1 (Very Low)
Not Assessed

Note: The indicators are numbered here to correspond with the Critical Success Criteria as specified in the National Action Plan: Regional Assessment Process (Version 1).

Critical Success Criteria Assessment

Assumptions

To determine regional progress toward successful coordination of Australia's clean energy and industry opportunity, the assessment draws on the Australian Energy Market Operator's (AEMO) Integrated Systems Plan. The plan has modelled pathways that support the retirement of coal-fired power generation in Australia and the modernisation of energy systems, primarily through the use of renewable energy. The AEMO scenario that best reflects the energy system requirements to build a clean export economy is AEMO's Green Energy Exports Scenario.¹ This scenario approaches an electricity grid that is net zero by 2035 and as such reflects the requirements of an energy system that is aligned with limiting the rise of global temperatures to within 1.5°C of global warming. As such, the Indicators^(Footnote 1) for Enabling Infrastructure (Energy System) are benchmarked against the AEMO Green Energy Exports scenario. Current system planning is taken to align with the AEMO Step Change scenario.²

Indicators in the Critical Success Criteria under the themes of Industry Capability and Capacity, and Policy and Governance, are assessed against a target of a 75% reduction in emissions by 2035 below 2005 levels, consistent with a trajectory to limit global warming to 1.5°C, as outlined in the IPCC's SSP1-1.9 scenario.^{3 4 5}

1.1 Enabling Infrastructure: Energy System

NSW is currently powered by 32% renewable energy.⁶

Given that current system planning is taken to align with the AEMO Step Change scenario,⁷ current energy system planning, including all current projects committed, is likely to deliver 77% renewable energy by 2035 if all projects in the pipeline are delivered. If the project pipeline is expanded to include uncommitted and anticipated network projects, the share of renewable energy in the fuel mix could rise up to 93%.⁸

An infrastructure build aimed at powering industry with renewable energy and enabling clean export opportunities—aligned with AEMO's Green Energy Export scenario—would see NSW powered by 81% renewable energy by 2035, without major new network projects. With the inclusion of currently uncommitted and anticipated projects, this share could rise to as much as 99%.⁹

These estimates come from the AEMO 2024 ISP generation and storage outlook data sets^{8,9} where the “counterfactual” cases illustrate results based on only currently committed projects. The Step Change and Green Energy Exports least-cost candidate development paths in these data sets indicate outcomes based on currently committed projects as well as the anticipated major new network projects

¹ The Critical Success Criteria can be broken down into five **Themes** (first level), 17 **Categories** (second level) and 55 **Indicators** (third level) - see Tables 1 and 2. **Definitions: Themes** represent the top-level conditions that underpin success—such as Enabling Infrastructure, Policy and Governance, and Social Acceptance. **Categories** are sub-areas within each theme, such as water infrastructure, workforce, or federal policy settings. **Indicators** are the specific, measurable criteria used to assess whether each category is ‘ready’ or needs attention. For more details, see [National Action Plan: Regional Assessment Process \(Version 1\)](#).

for these scenarios. The percentage of renewable energy generation takes into account the modelled energy generation as well as demand.

Timely development of three Renewable Energy Zones (REZs)—Hunter-Central Coast, New England, Central-West Orana—and the Hunter-Central Coast offshore wind zone (as defined by AEMO) is crucial to delivering sufficient firm renewable energy for NSW load centres and specifically the Hunter region industrial hubs and precincts (Muswellbrook Clean Industries Precinct, Hunter Energy Hub (AGL) and Port of Newcastle Clean Energy Precinct) (Figure 1 and Table 3).

Table 3: REZ Generation by GW for AEMO ISP Step Change and Green Energy Exports Scenarios

Transmission Project	Status (NSW Planning)
New England	11.3-12.7
Central-West Orana	10.4-14.8
Hunter-Central Coast	1.5-2
Hunter-Central Coast Offshore Wind	5.2 (Note current planned capacity is 2 GW ¹⁰)
Total generation capacity	28.6 - 34.7

Ref: AEMO, [DCCEEW](#)

The Hunter's three remaining coal-fired power generators (Bayswater, Eraring and Vales Point) are all scheduled to close by 2033, creating a new energy demand of 34 TWh p.a. across NSW¹⁰.^(Footnote 2) By 2033, the combination of the Hunter, New England and Central-West Orana REZs are projected to produce 48.5-59 TWh p.a., with production projected to grow to between 60-75.5 TWh p.a. by 2035.¹¹

Our analysis indicates that powering a Hunter Renewable Energy Industrial Precinct (REIP) with 100% renewable energy (both existing industry and new clean industry opportunities such as hydrogen) will require 22 GW,¹² equal to around 58 TWh p.a., after applying a 30% capacity factor.^(Footnotes 3,4)

The projected energy generation from regional REZs (Figure 1) has the potential to initiate industrial electrification. However, the anticipated growth in electricity demand—driven by building electrification and widespread electric vehicle adoption across NSW—will likely consume a significant proportion of this energy. Energy efficiency gains across these sectors and industry, along with activation of significant consumer/ distributed energy resources, will free up some of this generation for industrial electrification. However there is likely to remain a shortfall. Uncertainty around the amount of renewable energy available to industry has the potential to hamper industry decarbonisation capability, notably

² Total energy capacity of all three powerstation 6960 MW X hours per year 8760 X capacity factor 55% = 33,500 GWh

³ The ratio of the actual energy produced by an energy generation system over a period of time to its maximum possible output if it were running at full capacity for the same period. Capacity factor = Actual energy output divided by maximum possible energy output at full capacity.

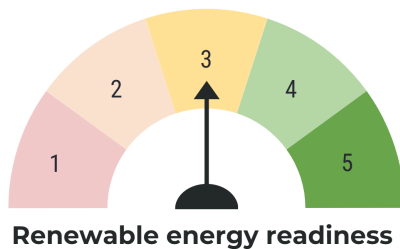
⁴ An updated assessment of the renewable energy required for a Hunter REIP will be a focus of future research by BZE.

where green hydrogen production is required to meet high heat processing needs.

Figure 1: Hunter REZs and Clean Energy Precincts



1.1.1 Renewable energy generation



Indicator grading scale

- 1 - Pipeline of renewable energy projects inadequate to meet more than 50% of REZ needs by 2035
- 2 - Pipeline of renewable energy projects adequate in line with 2035 Green Energy Exports targets, more than 75% without planning approval
- 3 - Pipeline of renewable energy projects adequate in line with 2035 Green Energy Exports targets, more than 50% without planning approval
- 4 - Pipeline of renewable energy projects adequate in line with 2035 Green Energy Exports targets, more than 25% without planning approval
- 5 - Pipeline of renewable energy projects adequate in line with 2035 Green Energy Exports targets, all projects approved to deliver REZ, at least 15% already under operation

Assessment

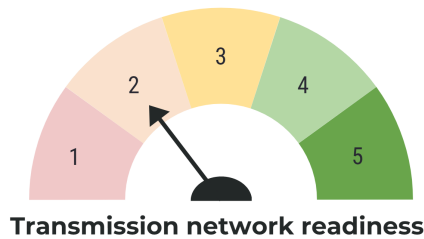
The deployment of renewable energy available to Hunter industry is ranked as 3, indicating significant progress, but will likely fall short of what is required to replace coal-fired power generation and enable clean industry opportunities by 2035. This takes into consideration the current timeframe for project approvals in NSW and the need for an increase in the total generation capacity of projects required.

Despite a promising pipeline of renewable energy projects, the slow pace of planning approvals and lengthy construction timelines introduce significant uncertainty regarding whether these projects will be completed in time to replace coal-fired power generation. If all projects in the approvals pipeline reach project completion by 2033, renewable energy deployment will be close to meeting the required levels of energy to decarbonise existing industry and enable clean industry opportunities. According to BZE's analysis of collated project data, 74% of project generation capacity is yet to receive planning approval.¹² With approvals taking between 5-8 years in NSW¹³, the timely deployment of these projects is uncertain. Of the projects approved, less than one third of total capacity is under construction.¹⁴

Bridging the Gap: Infrastructure, Community Aspirations and NSW's Energy Future

NSW renewable energy targets align more closely with AEMO's Step Change scenario and Federal 2030 emission reduction targets.⁷ Barriers to enabling infrastructure installation and social acceptance are being addressed in the NSW "whole of government" approach.¹⁵ Successful delivery of projects such as the Golden Highway upgrade (port to REZ) are needed to better enable transport of renewable energy infrastructure (e.g. wind turbine blades) to locations where it is being installed. It is also essential that housing, social services, and community infrastructure such as water¹⁶⁻¹⁸ align with the aspirations and needs of REZ communities to alleviate significant bottlenecks to renewable energy projects while encouraging greater social acceptance. Coordinating these outcomes will help NSW reach its 2030 targets and create significant momentum and buy-in to realise the Hunter REIP economic opportunity.

1.1.2 Transmission network



Indicator grading scale

- 1 - Transmission network tracking to meet ~25%-50% of demand to electrify existing industry by 2035
- 2 - Transmission network tracking to meet ~more than 50% of demand to electrify existing industry by 2035
- 3 - Transmission network tracking to meet electrification of existing industry and up to 20% additional demand by 2035
- 4 - Transmission network projects on track to meet 100% electrification of existing industry and between 20-50% of additional demand by 2035
- 5 - Transmission network on track to meet 100% transmission needs of existing and new entrants by 2035

Assessment

The progress of the transmission networks servicing the Hunter region is assessed as 2, as it is not tracking to enable high levels of industrial electrification and green exports by 2035.

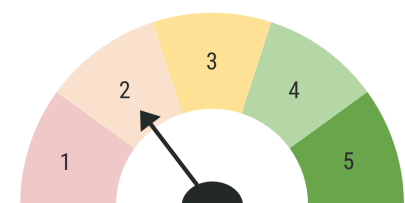
The current planned projects (tracking in line with AEMO's Step Change scenario) are insufficient to support large scale industrial electrification of existing industries in the Hunter. While current projects will come close to delivering the required capacity, many of these projects are in the early phase of development, with none in construction.¹⁴ Additionally, there is uncertainty around whether transmission projects can be delivered in a timeframe that enables renewable energy projects to connect as they reach completion. This has the potential to stifle investment in the required renewable energy generation and act as a deterrent to new clean industrial development in the region.

To deliver on the Green Energy Exports scenario, 16.5 GW of transmission capacity needs to be delivered by 2031. Currently, 90% of this capacity is scheduled to be delivered by 2031.¹⁴ NSW EnergyCo plans (Table 4) to build 15.5 GW of network capacity by 2031 across four projects¹⁴, noting that the Hunter-Central Coast REZ Network Infrastructure Project is a distribution focused REZ.¹⁹ Of the four projects, one has planning approval, two are in early stages of planning approvals, and one is under early design. One of the most critical projects, the Hunter Transmission project (HTP) is tracking to EnergyCo's timeline and is expected to begin construction in 2026. The HTP is considered one of the most critical transmission projects in the state and will form the backbone of the NSW electricity grid by unlocking renewable energy from the Central-West Orana and New England REZs. Additionally, it will service consumers across the Hunter, Sydney and Illawarra regions, which together consume 80% of the state's electricity.²⁰

Table 4: Delivery schedule of transmission projects connected to the Hunter REIP

Transmission Project	Additional Capacity (MW)	Completion (Green Energy Exports)	Schedule completion (EnergyCo)	Status (NSW Planning)
Central-West Orana REZ Network Infrastructure Project	4,500	August 2028	2028	Approved
Hunter Transmission Project (Sydney Ring North)	5,000	2028-29	2028	Prepare Environmental Impact Statement
New England REZ Network Infrastructure Project	5,000	2028-29	2031	Prepare Environmental Impact Statement
Hunter-Central Coast REZ Network Infrastructure Project	1,000	2029-30	2028	Design stage incl. community engagement Note: a distribution focused REZ
TOTAL	15.5 GW			

1.1.3 Distribution network



Distribution network readiness

Indicator grading scale

- 1 - No plans for electrification including at the household level
- 2 - Electrification planning for modest consumer energy resources (CER) uptake, not industry electrification
- 3 - Electrification planning for high uptake of CER, not industry electrification
- 4 - Electrification planning for 100% electrification of CER, and modest amount of industry electrification
- 5 - Electrification planning for 100% electrification of CER, and high amount of industry electrification and capacity for new entrants

Assessment

The Hunter energy distribution network is ranked as 2, signalling that it is not ready for industrial-scale electrification. While there is significant focus from the distribution network operator Ausgrid on consumer energy resources (e.g. household solar PV, electric vehicles and community batteries), there is insufficient publicly available evidence that the operator is planning for large-scale industrial electrification, or what level of industrial electrification is possible. Distribution networks connect the high voltage transmission network to the places where electricity is used, ‘distributing’ electricity at lower, more usable voltages.²¹ Distributed Network Service Providers (DNSP’s) require customers to submit an application for approval prior to making changes in how they use their energy. Without this

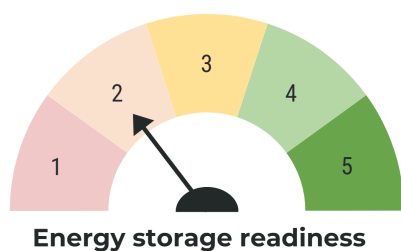
certainty for demand, it is hard for the DNSPs to include this potential increase in energy demand in their publishable forecasts.

Ausgrid is the determining authority and the DNSP for the Hunter Central Coast REZ (HCC REZ) project and as such does not require additional state planning approval,¹⁹ potentially avoiding long wait times associated with planning approvals. The HCC REZ is the first Distribution-REZ (DREZ) in Australia and is focused on augmentation of the distribution network to enable the efficient use and coordination of consumer energy resources across the network. Ausgrid operates within a regulatory framework that presents challenges in fully supporting the energy transition. To facilitate more effective operations, addressing specific constraints, such as the ring-fencing class waivers, would provide greater flexibility and better alignment with the evolving needs of the industry.

Given Ausgrid's coordinating role, there is potential for distribution to roll out at an increasing rate. Ausgrid has the opportunity to maximise existing available capacity of the distribution network and has commenced programs that address the changing energy distribution landscape driven by electrification of vehicles, homes and commercial buildings. Ausgrid is anticipating a doubling in solar customers, an 8-fold increase in home batteries, and nearly half a million EVs using the Ausgrid network by 2030.^{22,23} Further initiatives may be in development, but this information is not yet publicly available.

Planning for industrial electrification, however, is noticeably absent from Ausgrid's network planning with industrial energy distribution either not considered a constraint, or not an area of active consideration based on publicly available information. Without further information, the Hunter energy distribution network is assessed as not ready for industrial electrification at the necessary scale.

1.1.4 Energy storage



Indicator grading scale

- 1 - Pipeline of energy storage projects is less than 25% 2035 Green Energy Exports targets, more than 75% without planning approval
- 2 - Pipeline of energy storage projects is less than 50% of 2035 Green Energy Exports targets, more than 50% without planning approval
- 3 - Pipeline of energy storage projects adequate in line with 2035 Green Energy Exports targets, more than 25% without planning approval
- 4 - Pipeline of energy storage projects adequate in line with 2035 Green Energy Exports targets, all projects approved to deliver REZ, at least 15% already under operation
- 5 - Pipeline of energy storage projects adequate, in line with 2035 Green Energy Exports targets, more than 30% already under operation

Assessment

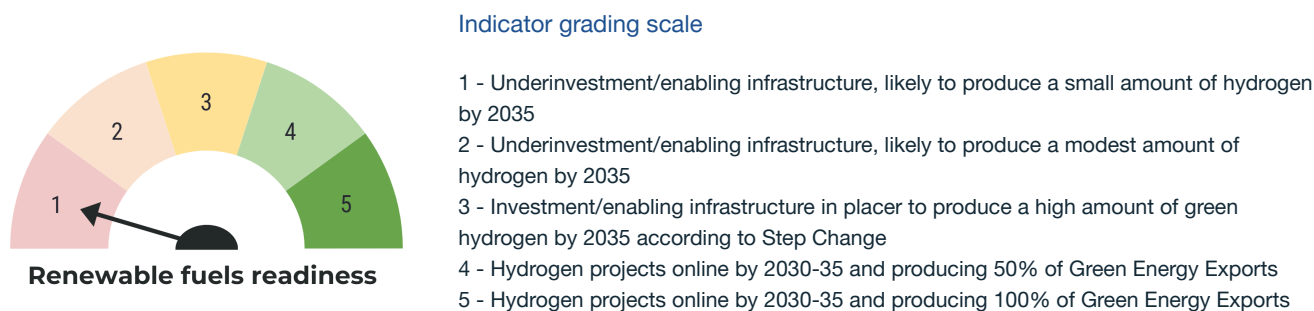
The existing pipeline of energy storage projects in NSW is assessed as a 2, tracking behind what is needed for NSW to deliver firming renewable energy in line with Green Energy Exports for the Hunter region.

AEMO modelling does not model the level of utility storage for each REZ, and instead works based on a statewide approach. NSW requires between 10.4-16.9 GW of utility-scale storage by 2035.^{9,24} BZE's

assessment (using project information sourced from Renew Economy²⁵ and collated project information from EnergyCo¹²) found that the current pipeline of utility-scale battery projects across the Hunter and neighbouring REZs (7.4 GW) covers 43% of utility storage needs outlined in the AEMO's Green Energy Exports scenario for NSW and 71% of requirements under AEMO's Step Change scenario.⁹ However, compared to the renewable energy pipeline, a significantly lower share of battery projects are awaiting planning approval. Currently, 45% of projects in the pipeline have received planning approvals, a further 16% are under construction, with around 40% of projects seeking planning approvals.¹²

A doubling of the pipeline of utility-scale battery projects is required in the Hunter to meet AEMO's Green Energy Exports modelling (17 GW), according to BZE's assessment. While the Australian Government's Capacity Investment Scheme (CIS) will support progress towards this goal, as of December 2024, the storage gap in NSW alone would effectively require the full capacity of the slated CIS storage (9 GW) to be filled. This underscores the urgent need to attract additional projects. Completion of all major transmission projects identified by AEMO could reduce the amount of utility-scale storage required, however, demand for storage would still exceed the current project pipeline (by at least 2 GW).

1.1.5 Hydrogen network



Assessment

Hydrogen projects have the potential to produce some hydrogen in the Hunter by 2035. However, given the limited number of projects currently in development, it is ranked as 1 as there is insufficient investment in necessary hydrogen infrastructure to meet anticipated demand and establish an industry capable of supplying industrial-scale green hydrogen in the Hunter.^{26,27} The Hunter Hydrogen Infrastructure Masterplan has modelled that by 2035, hydrogen demand could be as high as 1 Mt in NSW.²⁶ However, the current project pipeline, if delivered, can only deliver 245 kt of green hydrogen annually by 2035, leaving a 755 kt hydrogen shortfall.²⁶ This highlights a critical underinvestment in hydrogen and associated infrastructure.

The gap between projected demand and the green hydrogen project pipeline is expected to widen significantly following Fortescue Future Industries' withdrawal from AGL's Hunter Energy Hub²⁸ and Origin Energy's withdrawal of its interest in the Hunter Valley Hydrogen Hub in 2024.²⁹ Origin Energy has cited "uncertainty around the pace and timing of development of the hydrogen market", alongside the risk inherent in the capital-intensive venture,²⁹ when compared to renewables and energy storage as the factors behind its decision.

Key findings and recommendations

The Hunter region's required renewable energy to support the ongoing growth and sustenance of its industrial manufacturing is starting to come online. However, the pace of implementation of renewable energy infrastructure and the published ambition of renewable energy targets are not sufficient to deliver a prosperous, zero emissions future.

Pathways to industry decarbonisation require several key ingredients. For some manufacturers, electrification is the key process and providing affordable, firming renewable energy using solar, wind, storage (such as batteries and pumped hydro), transmission and distribution are the key elements of the renewable energy backbone that must be delivered. Additionally, for some industries (e.g. Orica's planned green ammonia production) green hydrogen and renewable fuels will also be required. The use of green hydrogen in industry decarbonisation will require technical and cost improvements in decarbonising affordable hydrogen supply chains alongside the firming renewable energy needs.

The project pipeline for renewable energy is strong, but timely project completion needs attention and faster approvals are needed to support clean exports and coal plant closures. Storage capacity may need to double to ensure sufficient firming energy. Transmission and distribution infrastructure are not currently on track to meet the needs of large-scale industrial electrification. Green hydrogen and future fuels development are falling behind projected demand.

1. Greater emission reduction ambition can drive industry decarbonisation and innovation.

- a. Federal and state government emission reduction commitments need to be more ambitious - for example, aligned with the AEMO Green Energy Export ISP scenario, to limit the rise of global temperature to relatively safe levels. Ambitious targets and a clear and accelerated path to a zero-emission economy will provide confidence to industry and private investors, while encouraging the build-out of infrastructure at the level required. While the current trajectory of project delivery is building momentum, greater ambition, and clear and coordinated targets in line with the Green Energy Exports scenario, will yield better long term economic outcomes.

2. Accelerated deployment of firming renewables is needed through streamlined approvals and improved coordination:

- a. There is a need for more streamlined approvals for renewable infrastructure. With 74% of projects yet to receive planning approval, a process that is currently taking 5-8 years and of the projects approved, less than one third are under construction.
- b. The pipeline of projects that deliver firming renewable energy (renewable energy and storage) requires tighter coordination with transmission rollout to ensure that projects can connect to the electricity network without delay.

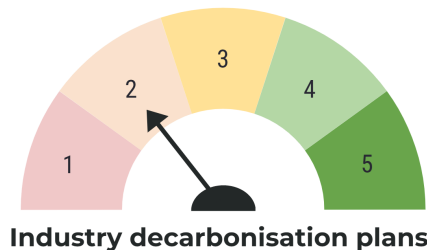
3. Regulatory reform is essential to manage and operate the distribution network more effectively during the energy transition.

- a. As the energy mix diversifies, DNSPs such as Ausgrid are increasingly required to manage more complex energy systems, including two-way flows. Updated regulation, such as removal of ring-fencing class waivers, would empower DNSPs to take a more

active role in supporting rapid decarbonisation. We recommend that these reforms be undertaken or led by the Australian Electricity Market Commission.

2.1 Industry Capability and Capacity: Industry decarbonisation

2.1.1 Industry decarbonisation plans



Indicator grading scale

- 1 - Most companies plans have plans with no targets and or not funded (Limited Plan/Insufficient Targets)
- 2 - Most companies have plans with targets, but only some are aligned to 75% emissions reduction by 2035 and not all commitments are funded (Basic Plans/Uncertain Implementation Pathways)
- 3 - Most companies have plans with targets, and most are aligned to 75% emissions reduction by 2035 and are funded (Intermediate Plans/Developing Momentum)
- 4 - All major companies have plans with targets aligned with 75% emissions reduction by 2035, with commitments funded but no internal performance initiatives (Comprehensive Plans/ Aligned with Regional Goals)
- 5 - All major companies have plans aligned with 75% emissions reduction by 2035, are fully funded and have internal performance initiatives (Leading Practice/Exemplary Industry Leadership)

Assessment

The Hunter's industrial decarbonisation plans were ranked as 2. This reflects that most energy-intensive manufacturing companies have decarbonisation plans with targets, with some aligned to 1.5°C (which we benchmark against 75% emissions reduction by 2035 below 2005 levels). However, most lack detailed decarbonisation pathways. While there is significant evidence of action, this still falls short of what is required to meet the necessary pace and scale of industry decarbonisation to limit the rise of global temperature to relatively safe levels (a 75% reduction in emissions by 2035). Decarbonisation plans are critical to reducing industry energy costs, maintaining competitiveness in global markets, and providing certainty of demand for renewable energy and hydrogen offtake.

Three of six energy-intensive manufacturers assessed have targets aligned with 75% emissions reduction by 2035 below 2005 levels:

- Tomago Aluminium, Australia's largest aluminium smelter, targets 100% renewable energy by 2035 by when it intends to have removed its scope 2 emissions (this is dependent on transmission being built in time.³⁰ Scope one emissions reduction requires anode technology, which is currently under development).
- Milltech Martin Bright, a steel bar manufacturer, has a Net Zero by 2030 target that it is progressing towards through electrification, energy efficiency, renewable energy and offsets.³¹
- Infrabuild, a steel manufacturer and recycler, has a target to achieve 76% reduction in emissions by 2030 by using renewable energy.³²

Orica, while not targeting 75% emissions reduction by 2035, has made significant progress to reduce emissions at its Hunter site, achieving significant emissions reduction through technology that eliminated 48% of greenhouse gas emissions in 2023.³³ It is also investing in the Hunter Hydrogen Hub with a view to replacing natural gas with green hydrogen and other low carbon options. However, the

timeline for green hydrogen has many dependencies such as cost, scale, available renewable energy, and investments, as well as research and development.

Collectively, the Hunter's heavy industry has made a significant start in implementing decarbonisation plans. However, there are key gaps, unclear implementation timelines, and technological breakthroughs required to achieve industry decarbonisation. Progress to date has depended heavily on government support. Greater transparency around plans and performance are anticipated through mandatory climate reporting, which will be required from January 2025.

Exclusion of fossil fuel companies

While coal and gas extraction companies were not included in this analysis, coal mines are significant emitters in the region. Their total emissions match those of the rest of the Hunter Region's industries. The majority of coal and gas extraction companies currently lack credible decarbonisation plans or evidence of action to reduce emissions. Inclusion of coal mining in the Hunter region's industrial decarbonisation evaluation would significantly lower the region's score.

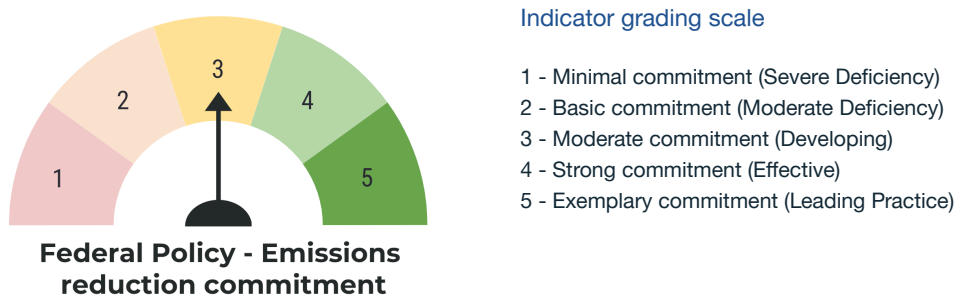
Key findings and recommendations

Most heavy industries have decarbonisation strategies but lack detailed pathways, with their success depending on timely infrastructure upgrades and affordable renewable energy. Greater collaboration between industry and project developers (e.g. through Purchase Power Agreements) can help provide certainty for firmed renewable energy. However, constraints in the supply of this energy to industry consumers needs to be resolved as described above. These agreements and other partnerships between industry and energy generators, and increased demand for them, can also provide a clear signal to governments regarding the industry support needed for greater ambition and project acceleration.

Industry decarbonisation requires access to affordable, reliable, firmed renewable energy, and there is significant government attention and focus on planning the required energy enabling infrastructure needed to repower manufacturing in the Hunter. However, as noted in the Enabling Infrastructure: Energy System section, the pace of deployment and coordination of energy infrastructure projects are significant challenges. This can have flow-on effects to influence investor confidence in industry decarbonisation.

4.1 Policy and Governance: Federal Policy

4.1.1 Federal Policy - Emissions reduction commitment



Assessment

The federal government's emissions reduction commitment is scored as 3, demonstrating a moderate commitment and reflecting progress but revealing the need for greater national coordination and more ambitious action. This indicator is an assessment of the national framework for emissions reduction. While it applies to the Hunter, it is not Hunter-specific.

The Climate Change Act 2022, while an improvement on previous commitments, is not aligned with the 1.5°C target. It sets a target of a 43% reduction in greenhouse gas emissions below 2005 levels by 2030 and net zero by 2050.^{34,35} The policy is more aligned with a 2°C pathway.³⁶⁻³⁹ It is worth noting that baselines matter. Targeting emissions reduction in Australia of 60% from the 2019 level by 2035 aligns with a target of 67% below 2005 levels by 2035.⁴ 75% emissions reduction by 2035 below 2005 levels is the broad benchmark used in this Critical Success Criteria assessment, with even greater emissions reduction commitment benchmarked for the electricity grid, i.e. what is needed to reach a zero-emissions grid by 2035 in line with the AEMO Green Energy Exports scenario.

The Safeguard Mechanism (SGM) is one of the key pieces of federal legislation designed to reduce Australia's industrial emissions. However, the SGM has loopholes that are a cause for concern. The SGM requires major emitters to reduce emissions by 4.9% annually.⁴⁰ However, trade-exposed baseline-adjusted (TEBA) facilities are permitted lower reduction rates, typically 2% annually, or as low as 1% in certain high-cost-impact manufacturing cases.⁴¹ The list of TEBA facilities is extensive across primary production facilities and includes mining, metal refinery, paper processing, ammonia, cement, and extracted oil and gas, to name a few.⁴² While this is touted to maintain competitiveness in international markets,⁴³ the slower pace of emissions reductions for TEBA facilities weakens the overall effectiveness of the mechanism, with the impact on competitiveness likely to diminish over time as global markets favour low and zero-emission alternatives.

Additionally, significant reliance on Australian Carbon Credit Units for offsetting emissions raises concerns about their ability to incentivise on-site emissions. While Australian offsets are some of the most rigorously assessed in the world,⁴⁴ uncurtailed reliance on offsets for emissions reduction remains

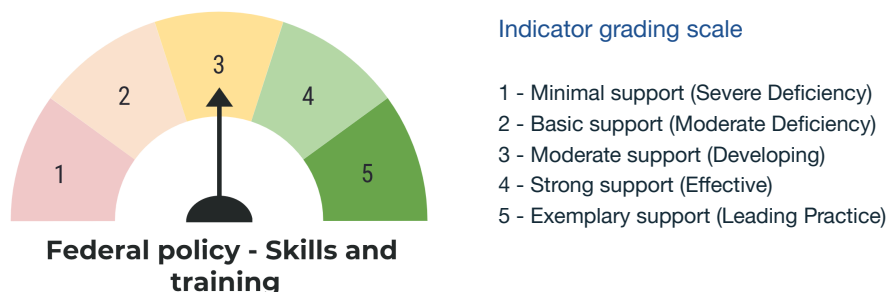
conceptually problematic because they have the potential to delay adoption of technologies to reduce on-site emissions in industrial processes.⁴⁵

The government's continued support for fossil fuels, particularly through the Future Gas Strategy, undermines its emissions reduction efforts.^{46,47} The strategy perpetuates the use of fossil fuels and locks in long-term fossil fuel infrastructure, conflicting with net-zero targets and clean manufacturing goals. Combined with coal mine expansion approvals and ongoing fossil fuel subsidies, these policies contrast sharply with global emission reduction targets and urgent calls from Pacific Island nations for stronger climate action, potentially undermining Australia's credibility in its bid to host Conference of the Parties (COP) 31.

The Future Gas Strategy undermines emission reduction efforts and risks locking in long-term fossil fuel infrastructure. It is also at odds with net-zero goals and contrasts with urgent calls from Pacific Island nations for stronger action.

To strengthen its emissions reduction strategy, the government could adopt more ambitious targets and reduce reliance on offsets to enhance direct emissions reduction measures, especially for trade-exposed industries. Aligning national policies with global climate science and phasing out fossil fuel investments would reinforce Australia's commitment to a zero-emissions future and position it as a credible leader in international climate action.

4.1.2 Federal Policy - Skills and training



Assessment

The federal government's skills and training policies are assessed as a 3 as they provide moderate support that addresses critical skills shortages for the energy transition. Key initiatives that are helping to drive this progress include micro-credential programs, skilled migrant visas, and investments in regional training infrastructure, such as the \$60 million Net Zero Manufacturing Centre of Excellence at Tighes Hill TAFE in Newcastle.⁴⁸ The recent establishment of the Net Zero Economy Authority (NZEa) provides a framework to support workers in select regions impacted by the energy transition, including the Hunter, to reskill for new jobs in clean energy industries.

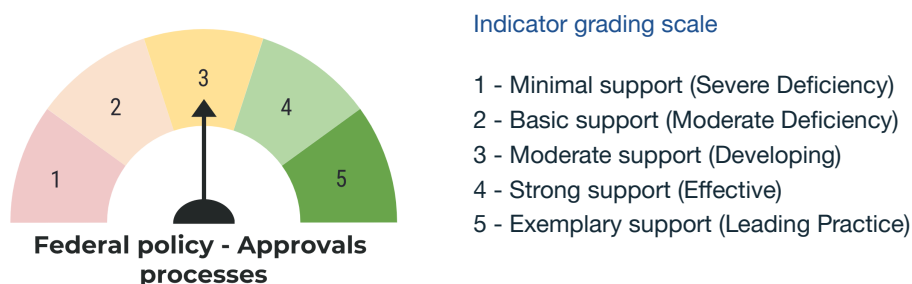
While significant progress is being made, some challenges persist, including the failure to adequately address skills gaps, barriers to workforce attraction in cleantech industries, and competition in the

labour market. The expansion of coal mines in the Hunter region^{49,50} may also impact local workforce availability, diverting workers from clean energy roles critical to the transition.

To address these challenges, reforms to improve workforce retention, expand training opportunities, and enhance the attractiveness of cleantech careers are essential.⁵¹ Leveraging the NZEA's strategic role will also be key to ensuring the Hunter Region meets its energy transition goals and supports Australia's clean energy future.

A nationally coordinated communications plan around skills for the energy transition will be key to ensuring a growing awareness of opportunities and skill pathways.⁵²

4.1.3 Federal Policy - Approvals processes



Assessment

Emissions will only fall when we replace fossil fuels with operational renewable energy in energy generation. [The federal government's approval processes are assessed as a 3, providing moderate support for the timely completion of projects.](#) This rating reflects moderate effectiveness, acknowledging progress in supporting renewable energy and infrastructure projects while highlighting ongoing challenges with extended timelines and procedural complexity.

The Environment Protection and Biodiversity Conservation Act 1999 (EPBC Act) requires projects that significantly impact Matters of National Environmental Significance (MNES) to undergo environmental assessments and approvals.⁵³ This includes major infrastructure, renewable energy, mining, and urban developments, often involving multi-step assessments across state and federal jurisdictions that can delay critical renewable energy projects.⁵⁴ The EPBC Act does not include a climate trigger, which means projects with significant impact on emissions can still be approved.

The regulatory landscape for land-use planning, infrastructure, and finance remains complex, with approval timeframes varying significantly across jurisdictions.^{47,55} Frequent regulatory changes, low inter-agency coordination, and consultation fatigue in communities impacted by concentrated renewable developments (alongside other developments) further exacerbate delays.⁵⁶ Stakeholders have advocated for a consistent national framework to synchronise approvals, reduce duplication, and streamline processes across government levels.⁵⁶

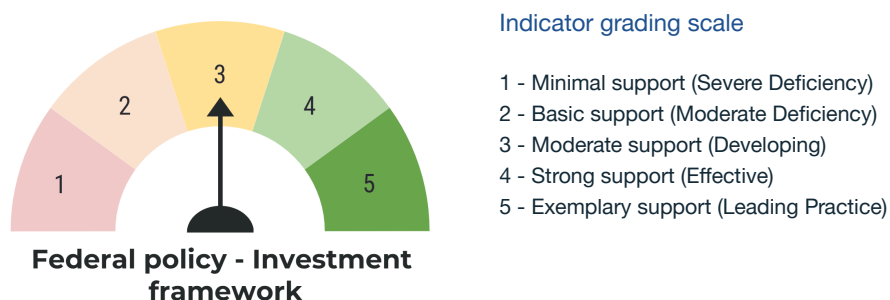
Recent initiatives to address these challenges include additional funding for the EPBC Act, and a single 'front door' for investors with leading practice investment proposals under the Future Made in Australia

(FMiA) framework. This single point of contact aims to simplify and coordinate approval processes for significant infrastructure projects, and attract global and domestic capital.⁵⁷ As of April 2025, consultations on the design of the ‘front door’ are nearing completion.⁵⁸ Although the FMiA Bill has passed, challenges remain if its provisions are not implemented effectively and on time.

Additionally, Infrastructure Australia and the Gateway Review Process play indirect but significant roles by evaluating project priorities and risks, ensuring alignment with long-term infrastructure needs, and mitigating delays through structured reviews. BZE’s REIP framework and the Climate Change Authority Sector Pathway Review recommend prioritising net-zero infrastructure projects. This prioritisation, combined with strategic land-use planning, could enable a more coordinated approach to approvals, reducing delays caused by fragmented processes and delivering key economic and energy infrastructure.

While initiatives like the “single front door” and increased funding signal progress, further regulatory reforms are essential to ensure projects are delivered in an efficient and timely manner. Streamlining overlapping federal, state, and local frameworks and improving inter-agency coordination will be critical to ensuring timely delivery of decarbonisation and renewable energy projects and other enabling infrastructure including roads and water infrastructure.

4.1.4 Federal Policy - Investment framework



Assessment

The federal government's investment frameworks are assessed as a 3, providing moderate support and reflecting progress in supporting the transition to a zero-emissions economy. While the government has introduced supportive initiatives, including the Clean Energy Finance Corporation, Australian Renewable Energy Agency (ARENA), the Capacity Investment Scheme, and the recent FMiA Act, the overall scope and impact of these investments remain limited given their recent introduction. A future assessment of these policies and applying criteria under the National Action Plan theme, Financing the Transition, will provide more insight into the adequacy of public investment and ability to catalyse private investment.

The FMiA Act 2024, guided by a National Interest Framework, prioritises investments in sectors with significant potential to contribute to net-zero goals. While the Act’s passage marks a significant step forward, its potential to catalyse private investment hinges on the effective and timely implementation of its programs.⁵⁹ Time will tell whether Australia’s policy signals attract globally competitive investment in cleantech.

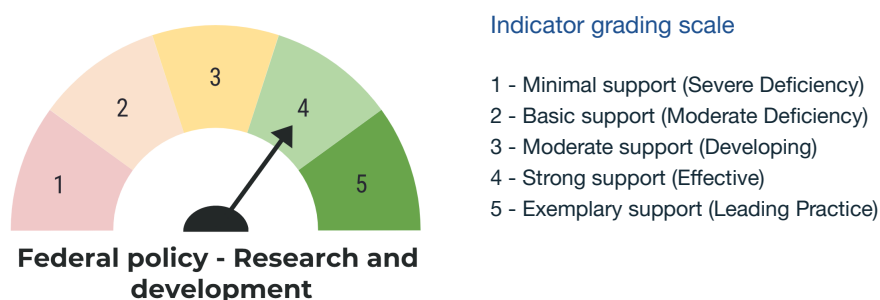
Key areas such as critical minerals processing and clean energy manufacturing have been earmarked for targeted funding under the FMI Act. However, Australia’s transition financing lags behind global competitors like the U.S. and EU, where large-scale incentives for clean energy industries have attracted significant private investment.⁶⁰ Additionally, investment coordination between federal and state governments remains a challenge (e.g. differing REZ locations, targets and deadlines for decarbonisation).⁶¹ Stronger cooperation frameworks between state and federal governments and consistent sectoral investment signals will improve the effectiveness of the investment landscape.

For the investment framework to achieve a 4 (“strong support” rating), the government will need to increase transparency, establish clear exit strategies from fossil fuel investments, and ensure the efficacy of future funding toward zero-emission industries. These actions could accelerate Australia’s pathway to net zero and strengthen its position as a global leader in the clean economy.

Preliminary findings indicate that while recent investments in clean energy and a zero-emissions economy are the largest in Australia’s history, they are being made alongside significant and ongoing support for fossil fuel extraction and retention in the energy system. This contradicting support of fossil fuel and zero-emission economies sends conflicting signals to investors and the energy sector, with the fossil fuel industry maintaining a distinct advantage in the absence of appropriate mechanisms to curtail expansion and mitigate exported emissions (Scope 3).

The interaction between the government’s investment framework and private investment will be examined in a forthcoming assessment under the Financing the Transition section of the Critical Success Criteria (see Table 1). This analysis will focus on aligning investment patterns to optimise the potential for a coordinated and effective transition to a zero-emissions economy.

4.1.5 Federal Policy - Research and development



Assessment

The federal government’s research and development (R&D) policies for energy transition are assessed as a 4, demonstrating strong support and reflecting substantial investment and strategic prioritisation. The 2024–25 Federal Budget demonstrates this commitment, allocating significant funding for R&D in clean energy and net-zero industries. Key initiatives include \$1.7 billion for the FMI Innovation Fund,⁶² a \$1 billion Solar Sunshot Initiative,⁶³ \$523.2 million for the Battery Breakthrough Initiative,⁶² and \$566 million to support critical minerals research.⁶⁴ Additionally, a \$6.7 billion tax incentive over ten years to

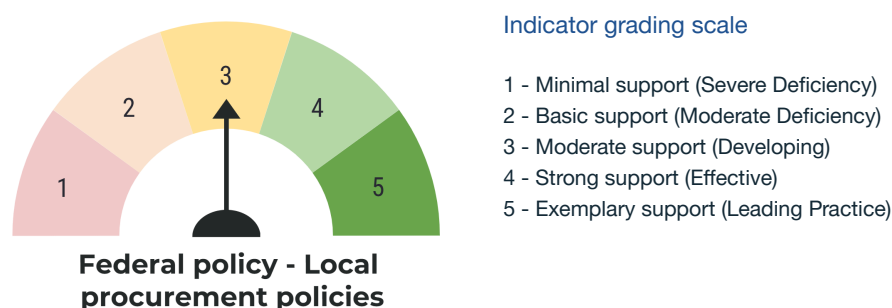
support hydrogen production underscores the government's focus on developing net-zero capabilities.⁶⁵

Historically, the federal government has allocated billions of dollars through the Clean Energy Finance Corporation and ARENA to support research and development in renewable energy and cleantech manufacturing.^{66,67} These initiatives, aligned with the Technology Investment Roadmap, continue to drive progress in areas such as renewable energy generation, hydrogen production, energy storage, and collaborative partnerships between industry and academia to facilitate the commercialisation of new technologies.

While the efforts noted above towards emissions reduction R&D are to be commended, there is an ongoing reliance on carbon capture and storage and carbon offsets, which risks overshadowing opportunities for direct emissions reduction solutions and technologies. To achieve leading practice (a 5 on the grading scale), greater focus is needed on direct emissions reduction strategies. This includes additional R&D funding and partnerships for hard-to-abate sectors such as agriculture, mining, and manufacturing, alongside enhanced support for the commercialisation and deployment of emerging technologies

The Climate Change Authority's Sector Pathways Review highlights the need for targeted R&D programs for hard-to-abate high-emission sectors.⁶⁸ BZE recommends challenge-based programs and concessional financing, which could both accelerate the development and adoption of these new decarbonisation technologies. A sector-specific approach, incorporating energy efficiency, electrification, and innovative practices, should aim to reduce reliance on carbon offsets as a primary solution and position Australia as a global leader in sustainable innovation.

4.1.6 Federal Policy - Local procurement policies



Assessment

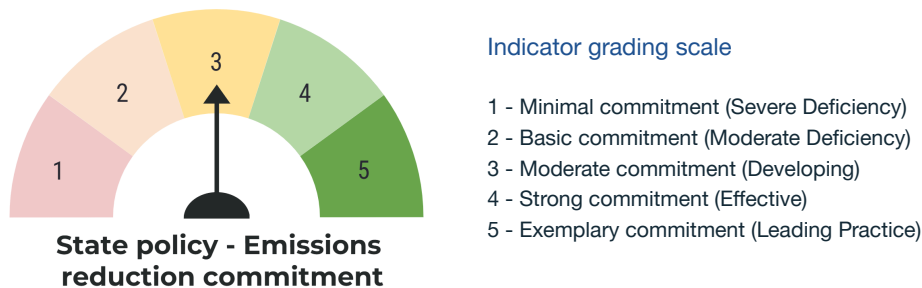
The federal government's procurement policies are assessed as a 3, demonstrating moderate support for the transition to a zero-emissions future. Recent initiatives, such as the Environmentally Sustainable Procurement Policy (ESPP) and the FMiA Act, highlight an increasing commitment to sustainable procurement practices. The ESPP mandates the prioritisation of low-emission, circular, and sustainable goods and services for contracts exceeding \$7.5 million.⁶⁹ Similarly, the FMiA Act and the updated Capacity Investment Scheme emphasise the inclusion of local content in renewable energy projects, aiming to strengthen domestic industries and create jobs.^{70,71}

While these initiatives represent significant progress, there is potential to amplify procurement's impact on zero-emissions goals. For instance, awarding additional points for local procurement in government tenders and offering higher-tier investment support for projects with substantial local content would incentivise sustainable suppliers.⁵² These measures could enhance investor and business confidence by creating stable demand for zero-emissions products, signalling the government's long-term commitment to market support.

As highlighted in BZE's *Make It Here: Building Australia's Clean Industrial Future* report, government programs currently focus heavily on "supply-side support" which helps build the ability to produce cleantech locally.⁵¹ However, increasing "demand-side support"—like using government purchases to create steady demand for these technologies—is just as important to support local industries. By strengthening this demand, the government could speed up the adoption of zero-emissions solutions and boost Australia's clean energy and technology sectors.⁵¹ Recent announcements of production tax credits for aluminium and other cleantech commodities, supported by the FMiA Act, are welcome. While these initiatives have significant potential to support local cleantech manufacturing, it is too early to assess their effectiveness, with outcomes dependent on successful implementation.

4.2 Policy and Governance: State Policy

4.2.1 State Policy - Emissions reduction commitment



Assessment

The NSW State Government's emissions reduction commitment is assessed as a 3 demonstrating moderate progress, supported by the legally binding targets set under the Climate Change (Net Zero Future) Act 2023, which was passed with multi-party support in November 2023. These targets include a 50% reduction by 2030, 70% by 2035, and net zero by 2050.⁷² The Act is complemented by an independent Net Zero Commission, tasked with monitoring progress and ensuring transparency.⁷²

While these targets reflect significant commitment, they fall short of what is needed to be done to limit global warming and are less ambitious than commitments made by other states and territories, such as South Australia and the Australian Capital Territory. The Net Zero Plan adopts a multi-sectoral approach, focusing on energy, transport, hydrogen, primary industries, and the built environment. This strategy is projected to create over 9,000 jobs and attract up to \$37 billion in private investment.⁷³ Financial support from the Climate Change Fund further bolsters emissions reduction efforts. However, enforceable sectoral policies and comprehensive regional support remain areas requiring improvement.⁷⁴

Ongoing approvals for fossil fuel developments, particularly for export, present an ongoing challenge.⁵⁰ These approvals risk undermining emissions reduction targets and by sending mixed signals to investors and labour markets, slow the state's transition to a zero-emissions economy. To strengthen its strategy, NSW could adopt more aggressive near-term targets, enhance enforcement mechanisms across all sectors, and align its actions more closely with the latest climate science recommendations. Restricting new fossil fuel developments and providing robust support for emerging zero-emissions technologies will be critical to improve the state's overall approach.

By addressing these gaps, NSW has the opportunity to position itself as a national leader in climate action, drive meaningful progress toward its net zero goals and deliver significant economic, social, and environmental benefits.

4.2.2 State Policy - Skills and training



Indicator grading scale

- 1 - Minimal support (Severe Deficiency)
- 2 - Basic support (Moderate Deficiency)
- 3 - Moderate support (Developing)
- 4 - Strong support (Effective)
- 5 - Exemplary support (Leading Practice)

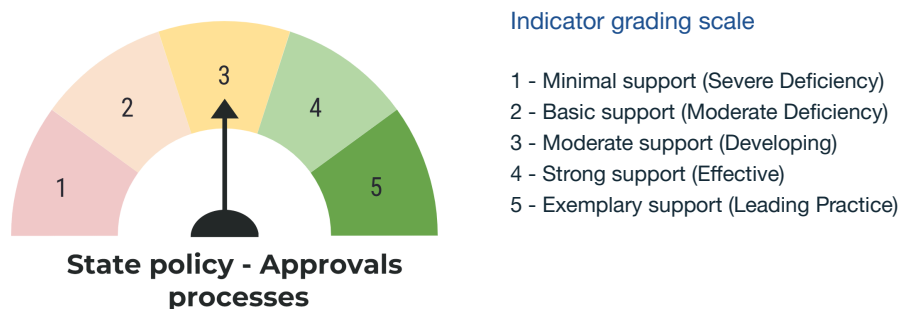
Assessment

The NSW State Government's skills and training policy is assessed as a 4, demonstrating strong support and reflecting a proactive approach to addressing the workforce needs of its "Renewable Energy Roadmap". The government's 2030 Renewable Energy Workforce Plan, currently under development, highlights a future-focused approach in tackling the skills challenges associated with the transition to renewable energy.⁷⁵ Projections indicate that the roadmap will create up to 7,000 new jobs, many in critical skill-shortage areas such as electricians, metal trades, and engineering. Of the 38 "critical occupations" identified in Jobs and Skills Australia's Clean Energy Capacity Study, approximately 28 are already in shortage, as outlined in the 2023 Skills Priority List.⁷⁶ Meeting the roadmap fluctuating job demand—estimated between 16,000 and 22,000 jobs annually—will require coordinated efforts to attract, reskill, and support workers over time.⁷⁷

The NSW government has committed to developing the Workforce Plan in consultation with industry, regional communities, and other stakeholders to ensure equitable skills and employment outcomes. This includes a focus on supporting underrepresented groups, such as First Nations Australians and women. Initiatives like the Renewable Energy Awareness and Career Training Centre (REACT) in Wellington⁷⁸ and the Net Zero Manufacturing TAFE Centre of Excellence in the Hunter⁷⁹ demonstrate this commitment, delivering targeted training to address regional skills shortages.

While these efforts underscore a robust approach to workforce development, further steps could elevate this policy arena to a leading practice level. Prioritising upskilling for workers transitioning from carbon-intensive sectors and strengthening pathways between vocational and higher education—particularly in engineering—would ensure the workforce is well-equipped for the demands of a net-zero economy. Enhanced collaboration across education and training providers, alongside industry-led initiatives, would further reinforce NSW's leadership in this space.

4.2.3 State Policy - Approvals processes



Assessment

The NSW State Government's approvals processes for renewable energy projects is assessed as a 3, demonstrating moderate alignment with zero-emissions goals, and supported by targeted measures to streamline approvals. Initiatives such as the establishment of REZs and a tiered approvals regime tailored to the scale and type of renewable energy systems reflect a commitment to expediting the development of clean energy infrastructure.⁸⁰ The government has also established an Independent Expert Advisory Panel for Energy Transition, which provides specialised knowledge to assess energy projects within these zones, improving decision-making and enhancing the efficiency of the approvals process.⁸¹

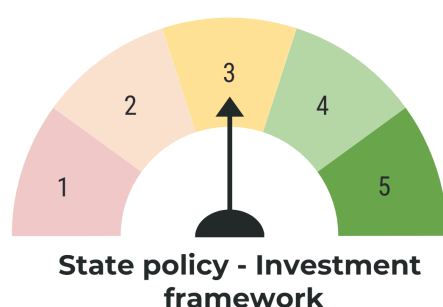
These efforts to streamline approval processes to reduce delays and support the efficient delivery of renewable energy projects are commendable. However, more needs to be done as challenges remain, with stakeholders raising concerns regarding regulatory hurdles and prolonged approval times for major clean energy developments.⁸² Between October 2018 and November 2023, the average approval times for projects were as follows:

- State Significant Development projects: 746 days
 - Wind projects: 3,488 days
 - Solar projects: 705 days
 - Battery projects: 530 days
- Critical State Significant Infrastructure projects (including hydro and transmission): 492 days⁸³

These timelines highlight the need for more efficient approval pathways to meet the pace required for building a zero-emissions future. Improving inter-agency coordination and introducing streamlined approval processes for zero-emissions technologies are critical steps. Additionally, applying consistent emissions reduction criteria across all sectors could reduce regulatory barriers and provide greater certainty for project developers.

Further reforms to improve timing certainty and reduce delays would significantly enhance NSW's ability to facilitate the rapid deployment of clean energy infrastructure. Such measures would ensure the state remains on track to achieve its ambitious renewable energy and emissions reduction targets.

4.2.4 State Policy - Investment framework



Indicator grading scale

- 1 - Minimal support (Severe Deficiency)
- 2 - Basic support (Moderate Deficiency)
- 3 - Moderate support (Developing)
- 4 - Strong support (Effective)
- 5 - Exemplary support (Leading Practice)

Assessment

The NSW State Government's investment framework is assessed as a 3, demonstrating moderate support for the zero-emissions transition, underpinned by two key initiatives.

- The Net Zero Plan Stage 1: 2020–2030 includes the Net Zero Industry and Innovation Program, which provides funding to support the development and deployment of clean technologies deployment and drive emissions reduction across key industry sectors.⁸⁴
- The Electricity Infrastructure Roadmap aims to attract private investment in 12 GW of renewable energy and 2 GW of long-duration storage by 2030.⁸⁵

These initiatives reflect welcome policy certainty and continuity in NSW around REZ development with the Net Zero Plan being implemented by the previous Liberal government and the current Labor government continuing to implement and build on it through the Electricity Infrastructure Roadmap.

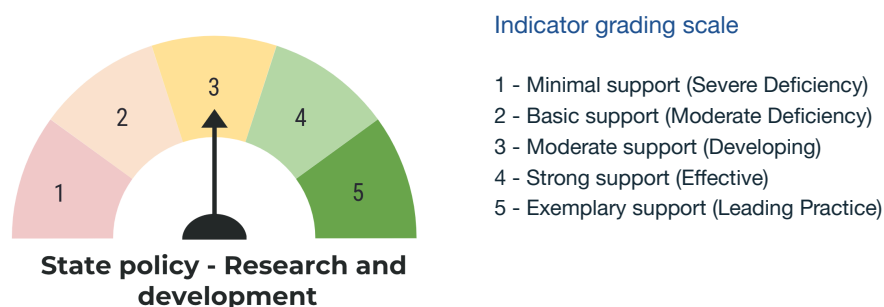
In September 2023, the NSW government committed \$1.8 billion to establish the Energy Security Corporation, accelerating the transition by improving grid connectivity and expediting grid connections for REZs. This funding prioritises early projects in the Central-West Orana REZ near Dubbo, projected to create 5,000 construction jobs and attract \$10 billion in investment by 2030.⁸⁶ Additional investments support critical projects such as the Hunter Transmission Project and the Waratah Super Battery, maintaining momentum across NSW's REZs.⁸⁶

To facilitate private sector engagement, Investment NSW collaborates with businesses to unlock investment, reduce barriers, and guide project planning. Resources such as the Large-Scale Solar Energy Guideline, Wind Energy Framework, and NSW Renewable Energy Map streamline project approvals and site selection for renewable energy projects, simplifying the development application process.⁸⁷

Additionally, the establishment of the Future Jobs and Investment Authorities underscores the government's commitment to help coal-reliant communities transition to future-focused, sustainable industries. These authorities, comprising local representatives, advise the government on regional priorities and investment opportunities, with the aim to deliver the best economic outcomes for these communities. They coordinate across all levels of government to ensure investment and action are timed appropriately to support coal-reliant regions.⁸⁸

This integrated approach highlights the government's dedication to fostering private investment and accelerating the renewable energy transition in NSW and is consistent with the aims of the federal Net Zero Economy Agency. Continued efforts to enhance funding, streamline approvals, and maintain strong private sector partnerships will be critical to scaling up investment and meeting the state's ambitious zero-emissions goals.

4.2.5 State Policy - Research and development

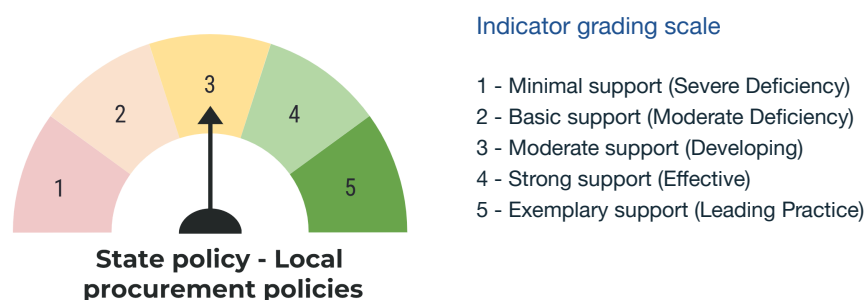


Assessment

The NSW State Government's R&D support for the zero-emissions transition is assessed as a 3, providing moderate support. Cleantech innovation was one of three focus areas within the Net Zero Industry and Innovation Program under the previous NSW government.⁸⁹ In 2024, as part of the Net Zero Manufacturing Initiative, the government launched a \$25 million Clean Technology Innovation Program to accelerate the research, development and commercialisation of emerging cleantech.⁹⁰ Additionally, the \$3.5 million Physical Sciences Fund supports the growth of innovative global companies headquartered in NSW, focusing on advancements in physical sciences and engineering with potential social, economic, or environmental benefits.⁹¹

The state has also established the Decarbonisation Innovation Hub in partnership with the University of New South Wales and the University of Newcastle, other leading universities, NSW government departments, and think tanks. The hub is intended to coordinate the efforts of government, industry and researchers to advance cleantech.⁹² The Decarb Hub is supporting the NSW Government's goal of net zero emissions by 2050 and a 50% reduction on 2005 levels by 2030.

4.2.6 State Policy - Local procurement policies



Assessment

The NSW State Government's local procurement policies are assessed as a 3, demonstrating moderate support for the zero-emissions transition, with a focus on sustainability and local sourcing. Key frameworks, such as the NSW Government Procurement Policy Framework and the Decarbonising Infrastructure Delivery Policy, incorporate environmental sustainability and embodied carbon measurement for major infrastructure projects.^{93,94} The establishment of the portfolio for the Minister for Domestic Manufacturing and Government Procurement on 28 September 2023, with The Hon. Courtney Houssos MLC as the inaugural minister, further reflects the state's commitment to enhancing sustainable procurement practices.⁹⁵

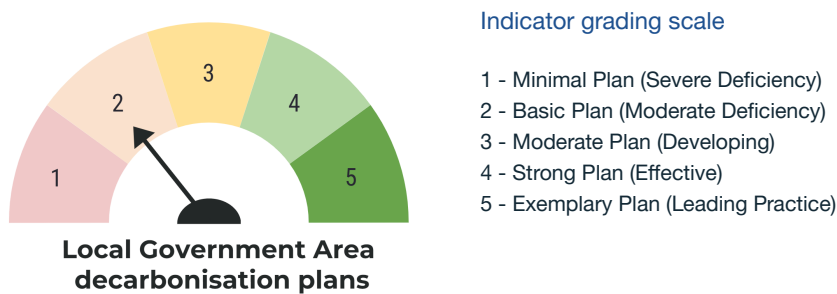
To strengthen local procurement, the government is introducing 'if not, why not' rules, requiring NSW Government agencies to prioritise local suppliers for projects exceeding \$7.5 million. If contracts are awarded to out-of-state suppliers, agencies must provide a detailed explanation, reinforcing the emphasis on local sourcing.⁹⁶

While these initiatives represent progress, there are opportunities to enhance alignment with zero-emissions goals. **Prioritising low-emission suppliers and suppliers with emission reduction commitments and introducing targeted incentives for zero-emissions practices would create stronger links between procurement policies and sustainability objectives.** Implementing enforceable guidelines that explicitly align with zero-emissions and regional development goals could further elevate the impact of NSW's procurement framework and bring it into closer alignment with the National Interest Framework.

By adopting these measures, NSW can strengthen its role as a leader in sustainable procurement, fostering regional economic growth while driving the transition to a zero-emissions future.

4.3 Local Government Policy

4.3 1 Local Government Area decarbonisation plans



Assessment

The Local Government Areas' (LGAs) decarbonisation plans within the Hunter region are assessed as **2**, reflecting early stage decarbonisation planning. There are 10 LGAs in the Hunter region,^{97,98} however MidCoast was excluded from assessment due to historical differences in community perception of the Hunter region's boundaries. Each LGA demonstrates varying levels of commitment to decarbonisation. Some councils have developed and are implementing ambitious strategies:

- City of Newcastle has a Climate Action Plan targeting net zero emissions by 2030.⁹⁹
- Singleton Council has pledged to source 100% of its electricity from renewable energy by 2030 and achieve 65% emissions reduction by 2030.¹⁰⁰

Other councils, however, do not have comprehensive frameworks that identify sector-specific actions and funding allocations to achieve outcomes.

To strengthen regional efforts, the state and federal government could support the LGAs in establishing clear and measurable emissions reduction targets across key sectors. Once plans are in place, this will need to be supported with resources for implementation. Enhanced stakeholder engagement (to be assessed at a later stage), with local industries and communities, would help ensure alignment with regional priorities and foster broader support for decarbonisation initiatives.

Improving monitoring and reporting mechanisms will also be critical for tracking progress and ensuring accountability. By adopting these measures, Hunter regional councils can create coordinated and actionable decarbonisation plans, making the best use of limited resources.

Key findings and recommendations

Federal and state energy and climate change policies are improving. However the suite of policies remains inconsistent and gives conflicting signals—such as promoting fossil fuel expansion alongside emissions reduction targets and clean industry development. Such conflicting signals to investors and the energy sector enable the fossil fuel industry to maintain a distinct advantage in the absence of appropriate mechanisms to curtail expansion. This risks diverting investment and skilled workers to a sector that should be in decline and away from the industries that will build future prosperity for the region.

Federal and state investment in meeting critical skill-shortages for the energy transition and new industries has received moderate to strong support through dedicated training facility investment and career pathway development. In general, policies at the federal, state and local level are supportive of future workplace requirements. However, as many of the plans are relatively nascent, their true impact will be seen in coming years.

A key piece of the puzzle missing is a national communications plan to attract workers to training opportunities and clear pathways to employment in the region. While governments can provide the former, the latter will need industry and local businesses to partner with governments to ensure that upskilled workers are in demand.

Delays in transitioning the Hunter to renewable-powered manufacturing risk competitiveness both in terms of energy costs and long-term economic stability, as global markets move away from coal and carbon-intensive commodities to clean energy and industries. As progress is made locally and coal/fossil fuel exports decline, it is crucial to accelerate efforts on the ground to develop the next economy. Establishing clear closure dates for fossil fuel assets would enable more effective workforce, economic and strategic land-use planning for the region, and allow private capital to move more effectively into renewable industries.

Most of the policies in this space are still new, meaning their full impact remains uncertain. Smaller players often struggle to navigate the complex policy landscape, with many either giving up or seizing the first available opportunity without the capacity to think more strategically. Small and medium enterprises, in particular, need targeted support, as they lack the resources of larger corporations to fully leverage policy incentives and transition effectively.

The Safeguard Mechanism, in its current form, permits fossil fuel and hard-to-abate industries to continue operating in a business-as-usual environment. While a review of the policy is scheduled for 2026, mandatory reporting requirements provide an opportunity for a midterm review aimed at introducing tighter guardrails. Strengthening the mechanism sooner would provide greater certainty for investors, industry, and communities preparing for the transition.

A clear direction on energy and economic policy is needed: the Hunter region will benefit from greater federal and state policy certainty around renewable energy targets, clean industry development, and associated incentives.

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